Let the Thermal Camera Decide When to Adapt: Infrared-Gated Test-Time Adaptation for Visible–Infrared Maritime Detection

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***Abstract*—Visible cameras at sea degrade in fog, low light, glare and sensor noise, and a detector trained on clean frames degrades with them. Re-estimating its batch-normalization statistics at test time needs no label, but adapting frames that were never shifted costs accuracy: up to 0.031 ship average precision (AP) at night on our development benchmark. A visible camera cannot tell a dark scene from a degraded sensor: its own health score flags every clean night frame as unhealthy. We let a co-mounted thermal camera decide. The method re-estimates the visible detector's batch-normalization statistics on eight-frame blocks only where the visible frame reports itself unhealthy and a healthy infrared frame reads daylight. It takes no gradient, changes no weight, never adapts the infrared detector, and has a causal form. On the Pohang Canal dataset, every variant was pre-registered before it was scored. The method passed development with no cell of 60 worse and replicated on a held-out run: +0.182 [+0.173, +0.192] ship AP under simulated sensor noise, +0.037 under fog, +0.019 under low light and, with a glare term, +0.031 under glare, with no cell worse. Inside a ten-pass ensemble its gains add, glare included, giving the best system we measured on every held-out cell with a corrupted visible stream. Earlier uses of the second sensor on the same system, as an uncertainty-weighted voter and as an image-statistic veto, did not survive registration. The corruptions are simulated, and the held-out run is a single daylight run.**

***Index Terms*—Batch normalization, infrared imaging, maritime object detection, sensor fusion, test-time adaptation, visible–infrared detection**

I. Introduction

Maritime perception systems increasingly pair a visible camera with a long-wave infrared (thermal) camera, on the assumption that the two fail in different conditions. Visible imagery degrades in fog, haze, glare, rain, darkness and sensor noise. Thermal imagery sees through darkness and glare but fails at thermal crossover, when vessel and water reach the same temperature. Neither failure announces itself. A visible detector trained on clean frames meets a fogged or underexposed frame with activation statistics its BatchNorm layers were not fitted to, and simply returns fewer and worse boxes.

Test-time adaptation of normalisation statistics is the cheapest repair available. Replacing or mixing the BatchNorm statistics with those of a test batch [1, 2] needs no label, no gradient and no change to any weight, and it recovers a large part of the loss under common corruptions. Its difficulty is not how to adapt but where. Adapting a frame that was never shifted moves a well-fitted detector away from its training statistics. In our screen, BN-mix applied to every frame gained on corrupted day frames but cost 0.007 ship AP on clean day frames and up to 0.031 on night frames (Section V-A). Selective test-time adaptation methods decide where to adapt from the model's own outputs or features: entropy, the divergence of one modality's prediction from the fused one, or feature statistics against a clean reference (Section II). A single visible camera faces an ambiguity here that none of these signals resolves. A dark frame may show a dark scene, which the detector was trained on, or a degraded camera, which it was not. The visible detector's own health score cannot tell them apart: on our development set it flags all 1,032 clean night frames as unhealthy and none of the 1,200 clean day frames (Section IV-A).

A thermal camera answers the question from a different physical process. It measures emitted radiation, so its reading of day and night does not share the visible camera's exposure, scattering or noise. Our method, IR-gated adaptation (registered as N4, the name we keep), re-estimates the visible detector's BatchNorm statistics only where three things hold: the visible frame reports itself unhealthy, the infrared frame reads daylight, and the infrared frame is itself healthy (Fig. 1). A dark or veiled visible frame that the thermal camera places in daylight is a degraded camera, not night, and it is adapted. A dark frame the thermal camera places at night is left alone.

We arrived at this design by elimination, on the same two-stream system, and the eliminations shaped it. The system was built so that predicted box uncertainty would weight the fusion of the two streams. A pre-registered test found that uncertainty informative about which boxes are right but not useful for fusing them (Section VI-A). An image-statistic veto that dropped the visible stream at night shipped instead, and it cost 0.1847 ship AP at night once the visible detector was retrained on corrected labels (Section VI-B). No learned gate we registered beat switching that veto off (Section VI-C). Through all of it the thermal camera proved a weak voter on boxes: on our held-out run its stream scores 0.0156 ship AP against 0.2767 for the visible stream. It proved a good judge of the scene.

Our contributions are:

1) **IR-gated test-time adaptation for visible–infrared detection.** A label- and gradient-free adaptation of the visible detector, gated per frame by the thermal camera's reading of illumination and by both streams' frame

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Appendices A–K, cited in the text, are provided as supplementary material.

health (Section IV). It has a causal form for live streams (N4c2), a glare term (N4g), and composes with a ten-pass ensemble (N4F). To our knowledge it is the first label- and gradient-free test-time adaptation for visible–infrared detection in which the second sensor gates adaptation by judging scene illumination (Section II).

2) **Pre-registered evidence, replicated on held-out data.** Every variant was registered before any of its numbers existed. Five passed development on 60 cells with no cell worse, one failed and was dropped, and the five that passed replicated on a held-out run: +0.1823 [+0.1733, +0.1916] ship AP under VIS sensor noise, +0.0370 under fog, +0.0186 under low light and +0.0309 under glare with the glare term, with no cell worse (Section V). Every scoring of the held-out run is logged in an exposure ledger.
3) **Negative results that locate where the second sensor helps.** On the same system, predicted uncertainty is informative but not useful for fusion; a sensor-selection veto encodes a claim about the detector and breaks when the detector changes; and frame-level gates pay a score-pooling penalty that a fixed ordering avoids (Section VI).

Adverse conditions are simulated, from a camera-chain model for fog, low light and glare and from ImageNet-C-style severities for noise, rain and blur. The held-out run is all daylight, and the gate does not fire on its clean frames, so no real degradation has tested the method (Section VII).

II. Related work

Test-time adaptation through normalisation. AdaBN [1] replaces BatchNorm statistics with those of the test batch. BN-mix [2] mixes test statistics with the source statistics at a prior strength N; we use it unchanged. TENT [3] trains the BatchNorm affine parameters by minimising entropy. Adapting where nothing shifted can hurt, and per-sample selection of what to adapt is established: SuMi [4] adapts only samples that pass feature-quartile and entropy tests, and updates normalisation affine parameters by gradient.

Multimodal test-time adaptation. Several methods adapt only the shifted modality, or reweight toward the reliable one. Each decides from the model's own predictions or features, and each updates parameters by gradient. READ [5] modulates cross-modal attention with a confidence-aware entropy loss. TSA [6] learns modality-specific adapters and a router that chooses which modality to adapt, and reports negative transfer when an unshifted modality is adapted, the failure our gate guards against. AdaPGC [7] flags the corrupted modality per sample by the divergence of its prediction from the fused prediction. CARAT [8] flags each sensor stream against its own clean statistics, and argues that cross-modal agreement misleads when sensors measure different physical processes. DASP [9] flags the biased modality by the redundancy of its fused features. ABPEM [10] bootstraps cross-attention from self-attention under a principal-entropy loss, and MM-TTA [11] adapts camera–LiDAR segmentation with cross-modal pseudo-labels. These are evaluated on audio–video classification, sentiment, image–text classification, wearable time series and segmentation. Night-TTA [12] adapts night colour–thermal segmentation by training BatchNorm affine parameters against an ensemble teacher, on every night frame.

Visible–infrared detection with illumination or uncertainty. UA-CMDet [13] weights each object's training loss by a rule built from cross-modal box overlap and RGB illumination, and at inference multiplies the RGB branch's scores by an illumination weight before non-maximum suppression. Zhao et al. [14] weight the visible and infrared branch losses by a label uncertainty estimated during training. Both use illumination or uncertainty to set fusion weights, not to adapt a detector. Gaussian YOLOv3 [15] attaches per-coordinate variance to a single-stage detector; we use its head (Section VI-A). Maritime YOLO variants such as RDSC-YOLOv4 [16] and YOLOv7-Sea [17] report accuracy without uncertainty or adaptation. The Pohang Canal dataset [18] with PoLaRIS annotations [19] provides paired visible–thermal maritime video with boxes on both streams; the Singapore Maritime Dataset [20] pairs visible with near-infrared video that need not show the same scene, and MassMIND [21] is thermal only.

What differs here. We found no test-time adaptation method for visible–infrared object detection; this rests on full-text reads of the eight closest papers (`docs/novelty-ttbn-2026-10-12.md`). Ours differs from the multimodal methods above in three ways. The thermal camera decides *whether* to adapt the visible detector, and it decides from a physical reading of the scene, not from agreement between model outputs. No gradient is taken and no weight changes. And every variant was registered before it was scored, and replicated on held-out data. Selective adaptation itself, adapting only the corrupted modality, is not our contribution, and neither is gating as such.

III. Setting

A. Data

The Pohang Canal dataset [18] records a 7.5 km route through canal, port and near-coastal water with a visible camera at 2048×1080 and a thermal camera at 640×512, both at 10 Hz. PoLaRIS [19] supplies ship and buoy boxes on both streams. There are five runs: pohang00 (day), pohang01 (night), and pohang02 to pohang04 (day). On disk they hold 158,319 images, 1,183,736 boxes and 28,388 paired visible–infrared frames. The two cameras are not co-registered: frames are paired by nearest timestamp, and the residual misalignment, measured, is 3–6 px median per run. Visible frames are letterboxed to 640×640; thermal frames are delivered as 8-bit images by per-frame min–max normalisation. Splits are interleaved contiguous blocks with guard bands, identical across modalities, and the night labels are the corrected set of a registered restore (Appendix A).

B. Detectors and fusion

Both streams use YOLO26-m [22]: the visible detector with two classes, the thermal detector with a single ship class and a P2 neck, because it cannot see buoys. The backbone was chosen on a 93-run benchmark, 31 YOLO variants from six

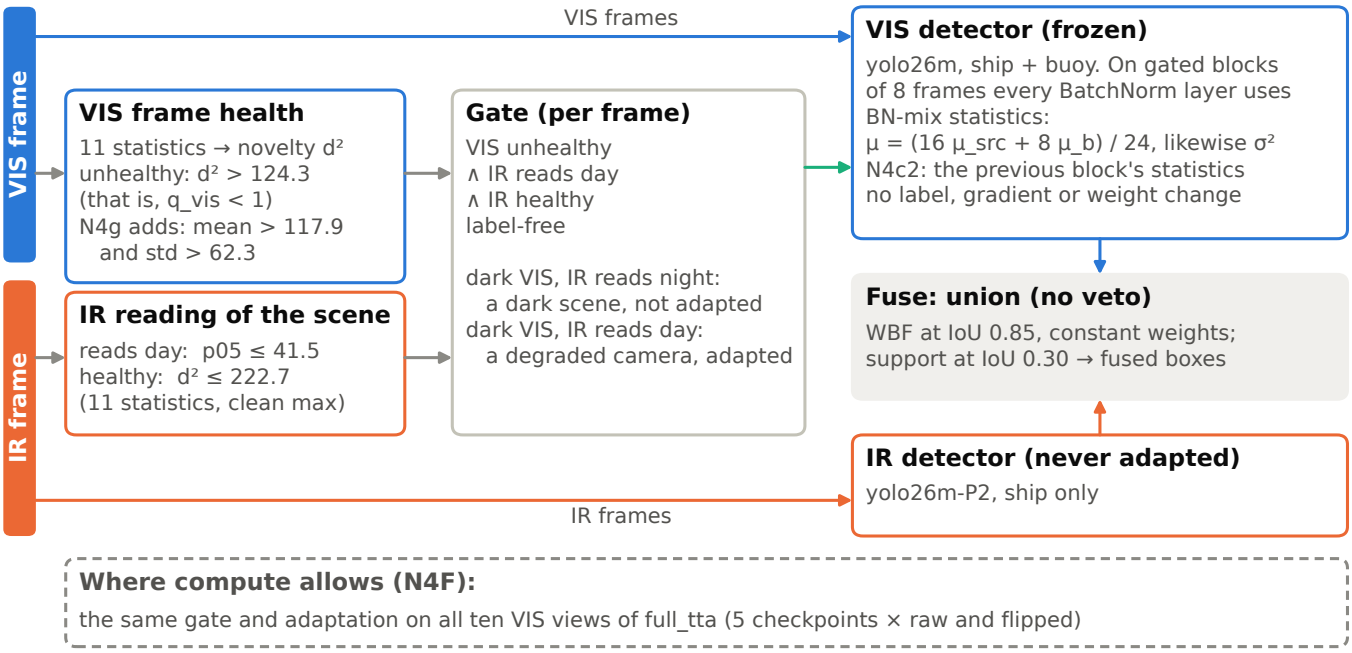


Fig. 1. **IR-gated test-time adaptation (N4).** Frame statistics from both cameras gate, per frame and without labels, whether the frozen visible detector normalises with BN-mix statistics re-estimated on a block of eight frames. The visible health score alone flags every clean night frame as unhealthy; the infrared day reading separates a dark scene from a degraded camera, and the infrared health bound keeps a fogged infrared stream from vouching for daylight. The infrared detector is never adapted, and the streams are fused by `union`, the shipped fusion with its veto switched off. Thresholds are the frozen constants the code loads (`runs/eval/structure_constants.json`; the glare term from `scripts/ttbn_variants.py`).

families with three seeds each, trained to a patience of 20 epochs (Table I; all 31 in Appendix H). The YOLO26 m, l and x variants form a top tier whose differences, at most 0.0040 mAP50-95, are below their seed standard deviations; the best variant of every other family sits at least 0.0142 below the leader. Within the tier, yolo26m is the fastest, at 57.0 frames per second per detector against 30.7 for yolo26x. We use five independently trained system pairs (visible seed k with thermal seed k), trained on the corrected labels with the held-out run removed from every list, and report the mean over the five.

The baseline fusion is `union`. Both streams pass to Weighted Boxes Fusion [23] at IoU 0.85 with constant weights (`w_vis` = 0.9926), and a box whose other-stream partner overlaps it at IoU 0.30 has its score raised by half (cross-modal support). At this threshold only 0.05 percent of visible boxes find a thermal partner, so the fusion is concatenation with a support bonus. The shipped system, `shipped` (preset `crossmodal26m`), is `union` plus a veto that drops the visible stream when the thermal camera votes night and the visible frame reads dark or veiled (Appendix D.2). A registered study of replacements for that veto found that switching it off, `union`, improves on it and that no gate we registered beats `union` (Section VI-C); `union` is therefore the baseline every adaptation is measured against. `full_tta` is the strongest system that changes no detector: five checkpoints per stream, each run on the raw and the horizontally flipped frame, merged per stream by WBF at IoU 0.55, with non-local-means denoising of thermal frames that fail their health check, and no veto. It costs ten detector passes per stream per frame.

C. Corruptions

Adverse conditions are simulated on the content rows of each frame. Fog, low light and glare follow a camera-chain model in physical order: atmosphere, optics, sensor noise, exposure, the dataset's own infrared stretch and quantisation. Fog is Koschmieder attenuation in linear radiance on per-frame metric depth [24] from Depth Anything V2 [25]. Low light is exposure scaling to the real night level, with the night noise measured on training frames. Glare is a source placed against the frame's computed horizon, with spread and exposure response fitted to the one real sun-entry event in training. Sensor noise, rain and motion blur are ImageNet-C-style severities [26] with no physical anchor. An earlier off-the-shelf corruption set was found broken at the source (its fog was a fixed 21-px blur) and retired; every result in the main body uses the camera-chain set (Appendix B).

D. Evaluation and statistics

Development. 2,232 paired validation frames, 1,200 by day (pohang00, 02, 03) and 1,032 by night (pohang01). The grid crosses five visible conditions (clean, fog at severities 1 and 2, low light, glare) with six thermal conditions (clean, fog at severities 1–3, noise at severities 2–3), by day and by night: 60 cells. Corrupted cells average three corruption draws per system; a cell's value is the mean over the five systems, with a 95 percent t-interval (df 4) on paired deltas.

Held-out. pohang04, an all-daylight run of 12,482 frame pairs, was never scored by any fusion system before a single pre-registered look, and none of the five system pairs or their reference statistics saw it. It has no thermal labels, so every

TABLE I
BACKBONE BENCHMARK: THE YOLO26 TIER AND THE BEST VARIANT OF EACH OTHER FAMILY

Variant	Family	Rank	mAP50-95	mAP50	FPS
yolo26x	YOLO26	1	0.2666 ± 0.0040	0.668 ± 0.033	30.7
yolo26l	YOLO26	2	0.2664 ± 0.0070	0.677 ± 0.003	44.9
yolo26m (used)	YOLO26	3	0.2626 ± 0.0109	0.672 ± 0.026	57.0
yolov10x	YOLOv10	4	0.2524 ± 0.0051	0.642 ± 0.010	34.9
yolo12x	YOLO12	5	0.2523 ± 0.0091	0.641 ± 0.017	23.8
yolov8l	YOLOv8	8	0.2470 ± 0.0033	0.634 ± 0.006	41.1
yolo11x	YOLO11	9	0.2466 ± 0.0078	0.650 ± 0.022	29.0
yolo26n	YOLO26	15	0.2384 ± 0.0064	0.653 ± 0.016	58.6
yolov9c	YOLOv9	16	0.2377 ± 0.0067	0.608 ± 0.018	43.6

Visible detector, 31 variants × 3 seeds trained on the corrected labels to a patience of 20 epochs; seed mean ± sd of the Ultralytics 8.4.90 validation mAP over ship and buoy. This is not local AP, so no value here is comparable with Tables II and III. *Rank* is among the 31 variants. *FPS* is one detector's batch-1 `predict()` at 640 × 640, fp32, GPU clock pinned at 1500 MHz, three seeds (two for yolo12x); two-stream throughput is half, before image statistics and fusion. Seeds whose continuation logs are missing score their 25-epoch best (14 of 93 runs, marked in the full table, Table S15). Sources: `docs/eval/bench_patience20_2026-10-08/table1_ext.md`, `phase1_benchmark/fps.csv`.

held-out number is scored against the visible ground truth (156,652 boxes). Nine cells corrupt one stream at a time with fresh draws (visible fog, low light, sensor noise, rain and blur with clean thermal; clean visible with thermal fog, noise and blur; and clean/clean), and a tenth, visible glare, was added for the glare term. Intervals are paired moving-block bootstraps (block 20 frames, 1,000 replicates), because frames at 10 Hz are autocorrelated and a frame-level bootstrap halves the true width (Appendix C.3).

Metric and rule. The metric is ship AP50-95 (local linear-interpolation AP), the only class both detectors emit. One rule decides every registered verdict, on development and held-out data alike: an arm passes (development) or replicates (held-out) if no cell is worse than its baseline by 0.0014 or more with the interval below zero, and at least one cell is better by 0.0060 or more with the interval above zero. The 0.0014 floor is the measured two-sigma noise of a paired delta; 0.0060 is the floor widened by the measured autocorrelation inflation (Appendix C.2–C.3).

Registration. Each arm's prediction, cells, statistic and rule were written and committed before any of its numbers existed. The held-out run fits nothing: no constant, threshold or calibration of any system. It is scored only through registered tests, and each scoring is logged in an exposure ledger. It has been scored nine times: the original look of the shipped system, a re-score of its corrupted cells after the corruption model was replaced, and seven later registrations, six of them the adaptation tests of Section V-B (Appendix G).

IV. METHOD

A. The ambiguity a single camera cannot resolve

A degraded visible frame raises a question its own pixels cannot answer: is the scene dark, or the camera? The visible health score `q_vis` is the clean-frame bound divided by the Mahalanobis novelty of eleven frame statistics (gradient Gini, the ratio of Laplacian to variance, spectral slope, texture cover and seven others), fitted on clean day frames of the development runs and capped at 1. It is a good detector of an unusual visible frame: it flags fogged, underexposed and noisy day frames. It is also, necessarily, a detector of night. On the development paired set it flags all 1,032 clean night frames as unhealthy and none of the 1,200 clean day frames (`runs/derived/structure/gauss_vis_paired_clean.json`, scored with `runs/eval/structure_constants.json`). Night frames are in the visible detector's training distribution, and adapting them hurts (Section V-A). The thermal camera's 5th-percentile reading separates the two cases: clean development day frames lie between 2 and 34 and clean night frames between 49 and 113 on its 8-bit scale.

B. The gate

Per frame, without labels, the gate fires when three conditions hold:

- the visible frame reports itself unhealthy: `q_vis` < 1;
- the thermal frame reads daylight: its 5th-percentile intensity `ir_p05` $\leq$ 41.5, the midpoint of the empty margin between clean day and clean night;
- the thermal frame is itself healthy: the Mahalanobis novelty of its own eleven frame statistics is within 222.7, the maximum over clean frames.

The third term exists because a degraded thermal stream can misread the scene: after the dataset's per-frame stretch, thermal fog and noise move a frame's 5th percentile across the day–night threshold (Appendix D.5), and without the health term the gate leaks onto up to 2.2 percent of night frames under thermal fog (`docs/prereg-ttbn-2026-10-11.md`). With it, the gate fires on no night frame under any thermal condition on development.

C. The adaptation

Frames are taken in non-overlapping blocks of eight consecutive frames. Where the gate fires, every BatchNorm layer of the visible detector normalises with the BN-mix statistics of Schneider et al. [2]:

$\mu = (N\mu_{\mathrm{src}} + n\mu_b)/(N+n)$, $\sigma^2 = (N\sigma^2_{\mathrm{src}} + n\sigma^2_b)/(N+n)$,

with N = 16, n = 8, and μ_b, σ^2_b the block's statistics over its frames and spatial positions, computed layer by layer as the block passes through the network. Frames the gate does not pass keep the detector's plain output. No gradient is taken, no weight changes, no label is read, and the thermal detector is never adapted; the two streams are then fused by `union`.

D. Variants

- **N4c2, causal.** N4 normalises a block with its own statistics, which include up to seven later frames (0.7 s at 10 Hz). N4c2 normalises each block with the statistics N4 measured on the previous block, at the cost of a second pass per block. For a live stream N4c2 is the version to use. A first causal variant, N4c, reused the statistics measured during the previous block's own adapted pass; it failed by construction (Section V-A).
- **N4g, glare.** N4's gate almost never fires under glare, because a glared frame is bright and textured and does not look novel to `q_vis`. N4g adds a visible-side term, mean intensity > 117.92 and standard deviation > 62.31, each the maximum over clean development day frames.
- **N4F, inside an ensemble.** The same gate and adaptation applied to every visible view of `full_tta` (ten passes per stream). N4Fg is N4F with N4g's gate.
- **N4ae, restoration on the same gate.** Gated frames that are dark (mean linear luminance below 0.18, auto-exposure's own target) take the detections of an auto-exposed frame instead.

Where the constants come from. N = 16 and the gate's terms were chosen on a screen draw before any development verdict; N = 64 halved both the gains and the losses. The visible health bound, the thermal day threshold and the thermal health bound are the frozen constants of the shipped system, fitted on clean development frames. Two choices used the held-out run's label-free frame statistics before any of its scores existed, and we disclose both. The thermal health term was switched from the shipped system's tighter authority bound (its 99th percentile) to the merge bound above after the former was seen to pass only 9.5 percent of clean held-out thermal frames (the latter passes 86.7 percent). The glare thresholds were checked to fire on 0.10 percent of clean held-out frames, after brightness percentiles that separate development glare perfectly were seen to fire on 55 percent of them. No constant was fitted on held-out labels.

V. Results

A. Development

Ungated adaptation hurts. In a screen on one corruption draw (two systems, not registered), BN-mix with N = 16 on every frame gained +0.0635 and +0.0767 ship AP on low-light day and +0.0088 and +0.0083 on fog by day, and cost 0.0080 and 0.0068 on clean day frames and up to 0.0311 at night (`docs/prereg-ttbn-2026-10-11.md`). The gate is designed to keep the first and remove the second.

N4 passes development (Table II). No cell of the 60 is worse than `union`; the worst Δ is -0.00001. Five are better beyond 0.0060 with intervals above zero: low light by day +0.0706 [+0.0593, +0.0818], low light with thermal fog at severity 1 +0.0357, fog at severity 1 +0.0094, fog +0.0091 and low light with thermal fog at severity 2 +0.0069. The gate fires on 16 cells, all by day: on 100 percent of low-light day frames and 99.6 percent of fogged day frames with clean thermal, and on no night frame under any thermal condition, so N4 equals `union` on every night cell. As thermal fog thickens the gate fires less, because the thermal stream stops vouching for daylight: on low-light day frames it fires on 42.5, 14.9 and 4.5 percent at thermal fog severities 1–3, and on none under thermal noise (`docs/eval/ttbn/ttbn_dev_2026-10-11.md`).

The variants. N4g gains +0.0240 under glare by day and +0.0130 with the thermal stream in light fog; its worst cell is -0.0008. N4c collapses from the first block that uses stale statistics, because they were measured under that block's own stale normalisation rather than N4's: mean maximum confidence falls to 0.21 against N4's 0.75, and it was not taken further. The corrected N4c2 passes; against N4 the price of causality is -0.0038 [-0.0045, -0.0031] on low-light day and within ± 0.0003 on fog. N4ae gains +0.0165 [+0.0053, +0.0278] over N4 on low light by day; darkness splits the conditions exactly, since every gated low-light frame is dark and no gated fog frame is, so on fog N4ae is N4. Inside the ensemble, N4F's gain survives almost undiminished (+0.0696 on low-light day, on its own draws), and so does N4g's glare gain (N4Fg, +0.0179 on glare by day). The held-out build runs in half precision with fused batch statistics; a development check priced that at no more than 0.0003.

B. Held-out replication

Each variant that passed development was registered for the held-out run before any of its held-out numbers existed. All six replicate (Table III, Fig. 2).

N4 (P-T): REPLICATES. No cell is worse than `union`, and three are better beyond 0.0060 with intervals above zero. Under visible sensor noise, re-estimating the normalisation statistics takes the fused output from 0.0169 to 0.1992; the ten-pass `full_tta` reaches only 0.0206 there. Fog gains +0.0370 and low light +0.0186. On the four cells with a clean visible stream the gate fires on no frame, so N4 is `union` there.

N4c2 (P-C2): REPLICATES. The causal version keeps N4's held-out gains, with no cell worse: causality costs at most 0.0007 against N4 (fog), less than the 0.0038 it cost on development low light.

N4g (P-G): REPLICATES. None of the nine registered cells corrupts the visible stream with glare, so its registration added a tenth. There N4g gains +0.0309 [+0.0290, +0.0327] over `union` (0.1417 to 0.1726), where N4 alone gains nothing; in the three other cells where the glare term adds two or three gated frames per draw, N4g equals N4 to the fourth decimal.

N4F (P-FT): REPLICATES. Inside the ensemble, adaptation adds what it adds alone, with no cell worse than `full_tta`: +0.2174 under noise (0.0206 to 0.2380), +0.0445 under fog (0.0586 to 0.1031), +0.0190 under low light and +0.0033 under rain. Under glare, N4's gate fires on 1.9 percent of frames and N4F equals `full_tta`, so glare was registered separately with N4g's gate (N4Fg, P-FG, after a development pass: +0.0179 on glare by day). It replicates: +0.0311 [+0.0286, +0.0332] over `full_tta` (0.1656 to 0.1967), +0.0240 above one-pass N4g, with no other cell worse than N4F. On each of the six held-out cells with a corrupted visible stream, N4F (N4Fg on glare) is the highest-scoring system we measured.

TABLE II
REGISTERED VARIANTS ON DEVELOPMENT: 60 CELLS, FIVE SYSTEMS, SHIP AP (LOCAL AP), PAIRED AGAINST THE BASELINE NAMED

Variant	Change	Baseline	Better	Worse	Largest gain [95% CI]	Verdict
N4	gate + BN-mix on 8-frame blocks	`union`	5	0	low light, day: +0.0706 [+0.0593, +0.0818]	PASS
N4g	N4 + glare term	`union`	2	0	glare, day: +0.0240 [+0.0178, +0.0302]	PASS
N4c	previous block's statistics, as measured in its adapted pass	`union`	0	10	low light, day: −0.1637	FAIL
N4c2	previous block's N4 statistics (causal)	`union`	4	0	low light, day: +0.0668 [+0.0559, +0.0777]	PASS
N4ae	N4 + auto-exposure on dark gated frames	`union`	5	0	low light, day: +0.0871 [+0.0734, +0.1008]	PASS
N4F	N4 on every view of `full_tta`	`full_tta`	5	0	low light, day: +0.0696 [+0.0620, +0.0757]	PASS
N4Fg	N4g's gate on every view of `full_tta`	`full_tta`	2	0	glare, day: +0.0179 [+0.0141, +0.0207]	PASS

Better / worse count cells beyond the rule's floors (Section III-D). N4F and N4Fg were scored on a second set of draws (VIS 945–947, IR 955–957), a disclosed second and third use; the others on VIS 942–944, IR 952–954. Intervals are 95 percent t-intervals over the five systems (df 4), except N4F's and N4Fg's, whose arms are five-system ensembles and whose intervals are moving-block over frames. N4Fg's rule had no gain requirement: no cell worse than `full_tta`, and none below N4F by 0.0014 where the glare term adds frames. Sources: `docs/eval/ttbn/ttbn_dev_2026-10-11.md`, `ttbn_variants_dev_2026-10-11.md`, `ttbn_causal2_dev_2026-10-11.md`, `ttbn_restore_dev_2026-10-11.md`, `ttbn_fulltta_dev_2026-10-11.md`, `ttbn_fulltta_glare_dev_2026-10-12.md`.

TABLE III
THE HELD-OUT RUN (POHANG04), ONE CORRUPTED STREAM PER CELL: SHIP AP (LOCAL AP) AGAINST THE VISIBLE GROUND TRUTH, MEAN OVER THE FIVE SYSTEMS AND THE FOUR DRAWS OF A CORRUPTED CELL

VIS condition (IR clean)	Gate	`union`	N4 − `union`	N4c2 − `union`	`full_tta`	N4F − `full_tta`
sensor noise	85.2%	0.0169	**+0.1823** [+0.1733, +0.1916]	+0.1820	0.0206	**+0.2174** [+0.2079, +0.2265]
fog	67.8%	0.0445	**+0.0370** [+0.0328, +0.0408]	+0.0363	0.0586	**+0.0445** [+0.0401, +0.0488]
glare (N4g, N4Fg)	81.0%	0.1417	**+0.0309** [+0.0290, +0.0327]	—	0.1656	**+0.0311** [+0.0286, +0.0332]
low light	85.2%	0.2152	**+0.0186** [+0.0154, +0.0219]	+0.0180	0.2458	**+0.0190** [+0.0152, +0.0232]
rain	18.6%	0.2209	+0.0036 [+0.0025, +0.0048]	+0.0035	0.2534	+0.0033 [+0.0022, +0.0047]
motion blur	9.2%	0.0419	+0.0001 [+0.0000, +0.0001]	+0.0001	0.0530	+0.0000 [+0.0000, +0.0001]
clean (IR clean, blurred, noisy or fogged)	0.0%	0.2635–0.2682	0	0	0.2923–0.2936	0

Δ is paired against the arm's own baseline (`union` for N4 and N4c2, `full_tta` for N4F) with a moving-block 95 percent interval (L = 20, 1,000 replicates). Gate = share of frames the gate passes. The glare row is N4g and N4Fg, whose gate fires on 81.0 percent of glare frames against N4's 1.9 percent (with N4's gate, N4F equals `full_tta` there). Sources: `docs/eval/ttbn/ttbn_p04_2026-10-11.md`, `ttbn_causal2_p04_2026-10-11.md`, `ttbn_fulltta_p04_2026-10-11.md`, `ttbn_glare_p04_2026-10-12.md`, `ttbn_fulltta_glare_p04_2026-10-12.md`.

N4ae (P-A): REPLICATES, with a cost development could not show. Low light gains +0.0259 [+0.0231, +0.0285] over `union`, +0.0073 [+0.0046, +0.0099] over N4, as on development. But the held-out rain darkens every gated rain frame below auto-exposure's threshold, so auto-exposure replaces adaptation there and N4ae on rain is 0.0036 below N4. Development has no rain. We do not recommend N4ae for that reason.

C. Where the gate does not fire

The gate's job is as much where it stays off as where it fires. It fires on no frame of the clean held-out cell, on a rounded 0.0 percent of the other clean-visible held-out cells, and on no development night frame under any thermal condition, so on those frames every variant equals its baseline. That matters because the baseline itself is right there. At night, `union` keeps the visible stream that the shipped veto discards. On a night test registered on pohang01's test split, 1,109 frame pairs that no fusion system had scored, `union` beats `shipped` by +0.0273 [+0.0154, +0.0400] and holds fused $\geq$ max(VIS, IR) at −0.0010 [−0.0049, +0.0031], while `shipped`, which vetoes the visible stream on 90.3 percent of those frames, fails it by 0.0283 (Appendix G.4). The detectors trained on the same run's training split, so this tests the fusion rule, not the detectors, and night AP levels swing from 0.04 to 0.35 between the run's 52-second episodes; directions, not levels, transfer.

D. An observation, not a test: ranking the adapted streams

N4 inherits a weakness of `union`. Where the visible stream is clean, the fused output falls below the visible stream alone, by 0.0085 to 0.0132 on the held-out run, because the thermal stream's boxes are interleaved with the visible stream's. A fixed ordering that ranks every visible box above every thermal box (`lex`) cannot fall below the visible stream: the scorer matches boxes greedily in score order, so a tail of boxes appended below every existing box cannot lower AP (Section VI-C). `ttbn_lex`, the same ordering of N4's adapted streams, is above N4 in all nine held-out cells, by +0.0096 to +0.0134 where the visible stream is clean and +0.0008 to +0.0067 where it is corrupted, and it holds fused $\geq$ max(VIS, IR) in every cell. No verdict was registered for it, and on development `lex` averaged 0.0037 below `union` over the 60 cells. The two data sets disagree, so we report it as a candidate for the next registration and recommend N4 with `union`, as registered.

VI. WHAT DID NOT WORK ON THE SAME SYSTEM

The adaptation was the last of a sequence of registered attempts to use the second sensor. The earlier ones are what

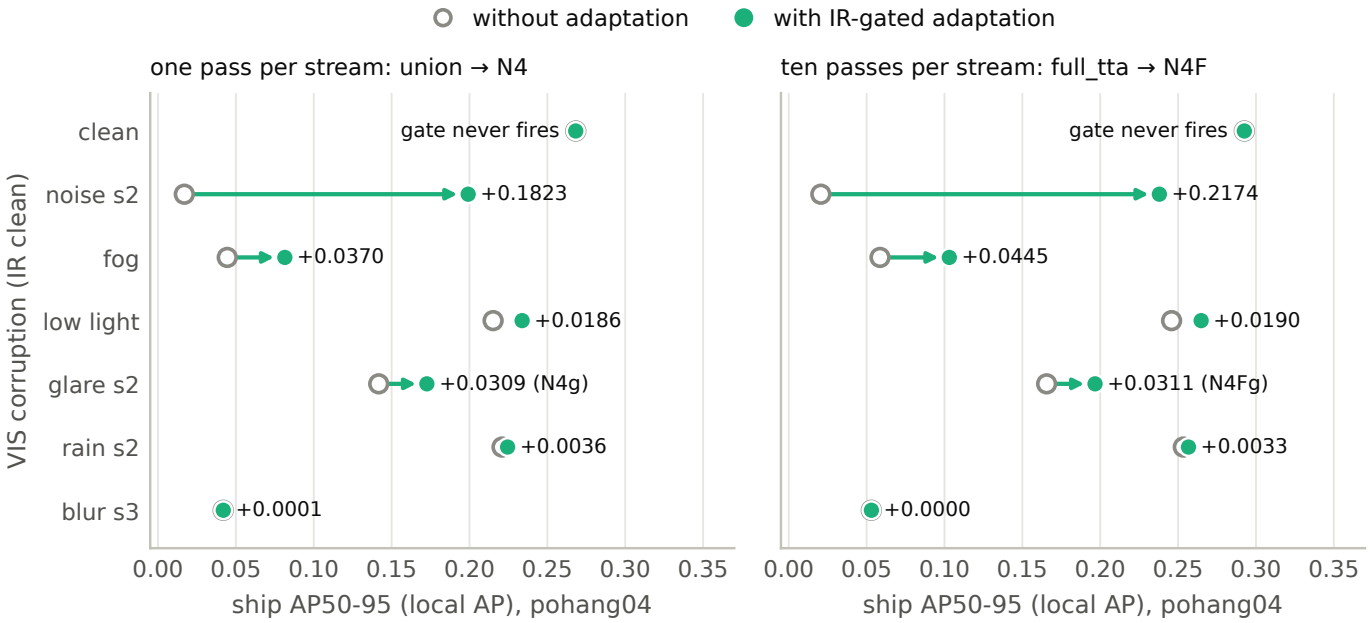


Fig. 2. **IR-gated test-time adaptation on the held-out run (pohang04), per cell.** Ship AP (local AP) against the visible ground truth, mean over the five systems and, in a corrupted cell, over four corruption draws; the thermal stream is clean in every row. Left, one pass per stream: `union` (hollow) and N4 (filled); the glare row shows N4g. Right, ten passes per stream: `full_tta` and N4F; the glare row shows N4Fg. Labels are the paired Δ; intervals are in Table III. On the clean cell the gate never fires, so the two markers coincide. Sources as Table III, with `docs/eval/holdout_p04_gatefree_2026-10-11.json` for the clean baselines; drawn by `scripts/paper_figures.py`.

made its design plain, and we summarise them here; the full record is in Appendices D–F.

A. Predicted uncertainty is informative but does not improve the fusion

Each detector carries a Gaussian variance head [15] trained with a beta-weighted negative log-likelihood [27], and the system was designed so that predicted σ would weight the fusion. We pre-registered the comparison that answers whether it should: real σ against the same σ shuffled onto the wrong boxes, at least three of four conditions above 0.0060 with a block-bootstrap interval excluding zero (Appendix E.2). Used as a weight on box coordinates, σ passes on none of four conditions, and the system's fusion weight was measured to take one value on every frame. Used to re-rank scores, real σ beats shuffled σ on three of four conditions, so it carries information about which boxes are right. But that information is within each detector: by day the gain is the visible detector re-ranking its own boxes, and adding the thermal stream moves it by -0.0040 to $+0.0035$ with no consistent sign. Against the system with no σ at all the re-ranked system is below on six of eight cells. A shuffled control establishes that a signal exists, not that using it helps. Learned jointly with confidence in a re-ranker of the visible boxes, σ adds +0.0050 [+0.0038, +0.0064] on one checkpoint, and its pre-registered replication on the five retrained detectors failed: positive on all five, beyond the floor on two. Two further lessons came from this test. The recorded runs had scored a ship-and-buoy macro instead of the registered ship AP, and the macro read NULL on the path where ship AP reads a pass; checking that the scored quantity is the registered one belongs in the registration. And on physically modelled corruptions the score-path pass falls to one or two of four.

B. A sensor veto is a claim about the detector

The system that shipped dropped the visible stream when the thermal camera voted night and the visible frame read dark or veiled (Appendix D.2). Its night arm was written while the visible detector scored 0.0000 at night. That zero turned out to be a label artifact: a filter had deleted every box in dark night frames from the training labels, and a registered restore of the boxes took night visible mAP50-95 from 0.0000 to 0.2520 (Appendix D.4). Retrained on the corrected labels, the visible detector sees at night, and the unchanged rule throws it away: on clean night the fused output equals the thermal stream alone and falls 0.1847 [0.1655, 0.2039] below the visible stream (Appendix D.3). Nothing in the image changed; what changed is the claim the rule makes about the detector. The rule gated on darkness, while what decides whether the visible stream should be dropped is its health, and with the retrained detector those variables no longer coincide. The same confusion appears on the thermal side: under a fogged or noisy thermal stream, the veto's fallback reads a dark low-light day as a fogged night and vetoes a working visible stream on up to 96 percent of day frames (Appendix D.5). The adaptation's gate keeps the two questions apart: it asks the thermal camera whether it is day, and each stream whether it is healthy.

C. No gate beat removing the gate

Seven learned or hand-written replacements for the veto were registered and scored on fresh corruption draws, from

per-stream calibration and Bayesian pair fusion to gradient-boosted trust over 32 backbone features (Appendix F). None beat switching the veto off: `union` is better than `shipped` on 30 of 60 development cells and worse on none, and every gate was worse than `shipped` somewhere by day. Two mechanisms explain most of it. A model of box correctness fitted to a noisy matching target re-orders a detector's own boxes worse than its confidence does. And a frame-level gate pools frames whose scores live on different scales: switching to the visible stream alone on a random three quarters of frames, a choice that carries no information, scores 0.0085 below the interpolation between the two pure policies. What worked used the scorer's structure. Boxes appended below every existing box cannot lower AP, so demoting a stream is never worse than deleting it, and a fixed tier order holds the floor of the better stream by construction (`lex`, Section V-D). The largest gain needed no fusion idea at all: ensembling frozen checkpoints helps every arm it is put under, and `full_tta` beats `shipped` on 57 of 60 development cells and on all nine held-out cells (Appendix G.3). The adaptation is the one change in this sequence that acts inside a detector, and it adds to the ensemble rather than competing with it.

VII. Discussion and limitations

The thermal camera is worth more as a judge than as a voter. As a second detector, the thermal stream adds little by day on this data: on the held-out run it scores 0.0156 against the visible stream's 0.2767, and fusing it lowers the output wherever the visible stream is clean (Section V-D). As a reading of the world it decided the largest gains we measured. Prior selective adaptation decides from the adapted model's own outputs or features (Section II). A second sensor that measures a different physical quantity gives a decision that does not share the first sensor's failure: an underexposed visible frame and a night frame look alike to every statistic of the visible frame and differ plainly in the thermal one.

What we recommend. For a two-stream detector built from frozen detectors: fuse by `union`, without a night veto; gate BN-mix adaptation of the visible detector on the thermal camera's daylight reading and both streams' health (N4), in its causal form (N4c2) for a live stream; and where ten detector passes per stream are affordable, apply the same adaptation inside an ensemble of frozen checkpoints with flipped views (N4F). Every element was registered before it was scored and replicated on the held-out run. Ranking visible boxes above thermal ones is a candidate, not a recommendation (Section V-D).

Limitations. The full list is in Appendix I; these bound the claim.

- *Simulated corruptions.* Every gain is on simulated corruptions. The largest, under sensor noise, is on an ImageNet-C-style severity with no physical anchor; fog, low light and glare follow a camera-chain model whose glare constants rest on one real sun event. The gate does not fire on clean held-out frames, so the method has not met a real degradation.
- *One held-out run, all daylight, scored repeatedly.* The held-out run is one daylight run, scored nine times through registered tests. Each later result is a repeated look at the same run, so agreement with development shows replication on one run, not generalisation across runs. Its thermal frames sit far from development's (median health novelty 106.4 against 7.9), which the merge bound tolerates; the night behaviour rests on development and on the pohang01 test split, which shares a run with the detectors' training data.
- *Label-free use of held-out statistics.* Two gate choices used the held-out run's frame statistics, without labels, before its scores existed (Section IV-D).
- *A healthy thermal stream is required.* Where the thermal stream is degraded the gate fires less, and every held-out adaptation cell has a clean thermal stream. A thermal camera at crossover was not modelled.
- *Development draws were reused.* Five development verdicts share one set of draws, and N4F and N4Fg reused the draws of the gate study; each later verdict is weaker evidence than a first test, as its registration states.
- *Cost.* N4 reads up to 0.7 s ahead; N4c2 removes that at a second pass per gated block. Neither N4 nor N4F was timed. The shipped two-stream pipeline runs at 9.1 FPS end to end on a laptop GPU, bound by its NumPy frame statistics (60.6 ms) rather than its two detectors (38.3 ms), and the gate reads the same statistics.
- *Dataset.* Pohang only, with one night run; the held-out ground truth is visible-only, so a true thermal-only detection counts as a false positive.

A note on the protocol. A measured paired noise floor, block-bootstrap intervals and magnitude floors moved 20 of 74 earlier findings of this project to indeterminate when applied retroactively, and a registration exposed a wrong metric in its own recorded runs (Section VI-A). The adaptation's gains survive these corrections by an order of magnitude: the smallest headline gain, +0.0186, is nearly eight times the 0.0024 maximum of the ship-AP noise floor (Appendix C).

VIII. Conclusion

Test-time adaptation of a visible detector is easy to apply and hard to aim: it helps on degraded frames and hurts on clean and night frames, and a visible camera cannot tell a dark scene from a dark sensor. A co-mounted thermal camera can. Gating BatchNorm re-estimation on the thermal camera's reading of daylight and on both streams' health keeps the gains and removes the losses: pre-registered, it passed development with no cell worse and replicated on a held-out run, +0.18 ship AP under sensor noise, +0.037 under fog, +0.031 under glare and +0.019 under low light, causal at a cost of at most 0.0007, and additive to a ten-pass ensemble. The same system's earlier attempts to use the second sensor, as an uncertainty-weighted voter and as a sensor veto, did not survive registration. The second sensor earned its place by judging the scene, not by voting on boxes.

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